

Battery Technology

The chemistry and engineering behind portable power

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Battery Technology develops the chemical and engineering systems that store electrical energy in portable, rechargeable form, powering devices from phones to electric vehicles.

DEFINITION

The study and development of devices that convert stored chemical energy into electrical energy through electrochemical reactions.

WHY IT MATTERS

Battery performance directly determines how far electric vehicles can travel, how long devices last, and how effectively renewable energy can be stored.

WORKING PRINCIPLE



Batteries store energy through reversible chemical reactions between electrodes and an electrolyte, releasing electrical current as ions move between electrodes during discharge, and reversing during charging.

QUICK FACTS

- ▶ The lithium-ion battery, now ubiquitous, was commercialized by Sony in 1991
- ▶ The 2019 Nobel Prize in Chemistry recognized the development of lithium-ion batteries

KEY TERMS

- Lithium-ion
- Electrochemical cell
- Energy density
- Battery management system

CORE CONCEPTS

- ▶ Electrochemical cells
- ▶ Battery chemistry (lithium-ion, etc.)
- ▶ Charge/discharge cycles
- ▶ Battery management systems

APPLICATIONS

- ▶ Electric vehicle battery packs
- ▶ Consumer electronics batteries
- ▶ Grid energy storage systems
- ▶ Medical device power supplies

ADVANTAGES

- ▶ Enables portable, rechargeable power
- ▶ Continuously improving energy density and lifespan
- ▶ Central to the shift toward electric transportation

CAREER OPPORTUNITIES

Battery research scientist, Electrochemical engineer, Battery systems engineer, Materials scientist

RESEARCH AREAS

Solid-state battery technology, Fast-charging battery design, Sustainable battery recycling

REAL-LIFE EXAMPLES

- ▶ Lithium-ion batteries in smartphones and laptops
- ▶ Electric vehicle battery packs
- ▶ Rechargeable household batteries

LIMITATIONS

- ▶ Limited by current chemistry's energy density ceiling
- ▶ Raw material extraction (lithium, cobalt) has environmental impact
- ▶ Degrades over repeated charge cycles

INDUSTRY APPLICATIONS

Automotive (EVs), Consumer electronics, Renewable energy storage, Aerospace

FUTURE SCOPE

Solid-state and next-generation battery chemistries promise significant improvements in range, safety, and charging speed.

CURRENT TRENDS

Major automakers and battery manufacturers racing to commercialize solid-state batteries for longer range and faster charging.

SUMMARY

Battery technology develops the rechargeable power systems behind phones, laptops, and electric vehicles, with rapid innovation ongoing.